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ML Assignment - 4

Q1) Hidden condition for k-Fold CV Misleading & Better validation Design

Hidden Condition:

k fold cross validation can be misleading if there is data leakage or temporal dependencies present in dataset. For ex., in time series data or data with sequential pattern, shuffling the data during cross-validation could mix data points that should not be mixed. This can lead to overly optimistic performance estimates during validation.

Better validation design:

For time series data, it's better to use Time series Cross-Validation where the training set is built on past data and the validation set contains data ~~set~~ that is temporally after the training data. This ensures that no further info leak into model during validation.

Q2) Why does Random Forest Outperform Plain Bagging?

- Both bagging & R.F are ensemble methods that uses multiple decision trees to reduce variance & improve model stability. R.F Typically outperforms plain Bagging because it adds a level of randomness during the construction of each tree.

i) Bagging : In bagging, each tree is built ~~using~~ ^{on} random sample of data ~~and~~ all features are considered when splitting at each node.

ii) Random Forest : In contrast of RF introduces additional level of randomness. At each split in DT, random subset of features is considered rather than using all features.

- The Random Forest combination with Bootstrap Dataset with feature randomness makes it less likely to ~~over~~ ^{overfit} compared to simple Bagging.